

The Mind's Eye as Interior Domain Physics:

Mental Imagery, the FSR Spectrum, and the Biology of Interior Experience

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Author Note

Nickolas Patrick Joseph Schoff, Schoff Research Program. This paper develops the biological and physical account of mental imagery as a bridge between established neuroscience and the Field State Representation (FSR) spectrum hypothesis of the Schoff Research Program. It draws on the Dimension-W framework (Schoff, 2025), the FSR Spectrum Hypothesis (Schoff & Claude, 2026a), the Invariant Agency framework (Schoff, 2026b), and the established neuroscience of visual imagery. The paper is designed to be self-contained. Correspondence: kiba3030@gmail.com. Archived at <https://doi.org/10.5281/zenodo.18989736>

Abstract

When a person closes their eyes and visualizes an image, they are performing something physically precise: generating top-down activation patterns in primary visual cortex (V1) that are functionally equivalent to the bottom-up patterns produced by actual visual input. The mind's eye is not metaphorical. It is the same neural hardware that processes external vision, driven in reverse. This paper develops the full account of what mental imagery is — biologically, experientially, and physically — and uses that account to ground the Schoff Research Program's FSR Spectrum Hypothesis in concrete neuroscience. The argument proceeds in three stages. First, we establish the neuroscience of mental imagery: the retinotopic organization of V1 imagery activation, the top-down generative pathway from prefrontal and parietal cortex through higher visual areas to V1, and the thalamocortical gating mechanism that normally maintains the boundary between internally generated and externally received visual representations. Second, we connect this neuroscience to the FSR Spectrum Hypothesis — which proposes that the Field State Representation is a continuous spectrum from maximally collapsed, intersubjectively convergent external reality at one pole to high-Freedom-Quanta interior imagination-space at the other — by identifying the thalamocortical gate as the physical implementation of the FSR boundary. When the gate is functioning, the two domains are appropriately segregated. When it fails — as in psychosis, hypnagogia, or extreme meditative absorption — the boundary becomes permeable and interior-domain experience becomes phenomenologically indistinguishable from exterior-domain experience. Third, we connect both to the Dimension-W framework, proposing that inward W-traversal is the physical correlate of the top-down generative pathway: visualization is not merely a neural process but an actual traversal inward along the W-dimension, accessing the constraint configurations stored in the informational remainder R. This account generates specific empirical predictions about the relationship between visualization precision, EC-depth, W-fiber coupling strength, and physiological outcome — connecting mental imagery science to the therapeutic predictions of the Constraint Topology Medicine framework.

Keywords: mental imagery, visual cortex, top-down processing, FSR spectrum, interior domain, thalamocortical gating, Dimension-W, visualization, psychosis, constraint topology medicine, informational remainder

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I. Introduction: The Question

When you look at a television screen, you know precisely what you are looking at. Liquid crystal or OLED pixels arranged in a grid emit photons in spatially organized patterns. Those photons reach your retina, activate photoreceptors in retinotopic order, and the signal propagates through your visual system, ultimately producing the experience of seeing an image. Every component of this process is physically characterized.

When you close your eyes and visualize — when you see an image in your mind's eye — what, precisely, are you looking at? The experience is phenomenologically similar to external vision: the image has spatial extent, color, motion, detail. It feels like seeing. But what is the physical substrate of that seeing? What is the display? What is the signal source? What is the screen?

This paper answers that question from neuroscience, and then asks what the answer implies for the physics of interior experience. The answer from neuroscience is precise and somewhat surprising: you are looking at the same neural hardware that processes external vision — primary visual cortex (V1) — but driven by a signal traveling in the opposite direction. The mind's eye is not a separate system. It is the visual system running backwards.

What this implies for the Schoff Research Program's theoretical framework is significant. The FSR Spectrum Hypothesis (Schoff & Claude, 2026a) proposes that interior imagination-space is physically real from the observer's frame — not less real than external reality, but differently real, operating at the high-FQ end of the Field State Representation spectrum. The neuroscience of mental imagery provides the biological mechanism that makes this theoretical claim concrete: interior experience is not floating free of physical substrate. It is V1 firing patterns generated top-down, with the same neural architecture and the same spatial organization as bottom-up visual processing.

The paper is organized as follows. Section II develops the neuroscience of mental imagery in detail. Section III connects this to the FSR Spectrum Hypothesis, identifying the thalamocortical gating system as the physical implementation of the FSR boundary. Section IV develops the Dimension-W account of visualization as inward W-traversal. Section V addresses the boundary failure conditions — psychosis, hypnagogia, shamanic states — where the gate opens and interior becomes exterior. Section VI develops the therapeutic implications for the CTM framework. Section VII presents empirical predictions.

II. What Mental Imagery Is: The Neuroscience

The Reverse Hierarchy: Vision Running Backwards

The visual system processes information through a hierarchical pathway. In normal bottom-up vision, the pathway runs: retina → lateral geniculate nucleus (LGN) of the thalamus → primary visual cortex (V1) → V2 → V4 → inferior temporal cortex (object recognition) → parietal cortex (spatial processing). Information flows from the periphery inward and upward, with each stage extracting increasingly abstract features: V1 responds to oriented edges; V4 to color and form; inferior temporal cortex to faces and objects.

Mental imagery reverses this pathway. Stephen Kosslyn's foundational research demonstrated through neuroimaging and transcranial magnetic stimulation (TMS) studies that V1 is genuinely activated during mental imagery — not in a reduced or approximate way, but with spatial organization that mirrors the imagined content (Kosslyn et al., 1999, 2001). When subjects imagine large objects, the activation spreads further across V1's retinotopic map — the spatial map that directly corresponds to the visual field — than when they imagine small objects. The spatial geometry of V1 activation during imagery is isomorphic to the spatial geometry of the imagined scene, just as it is during actual vision.

The signal driving this activation flows top-down: from frontal and parietal regions involved in working memory and attention, through the higher visual areas, back down to V1. This top-down generative pathway is not merely modulatory — it is the primary driver of the imagery experience. Disrupting V1 activity with TMS during mental imagery disrupts the imagery itself, demonstrating that V1 activation is not epiphenomenal but constitutive of the visual imagery experience (Kosslyn et al., 1999).

The Display: V1 as Shared Substrate

To answer the original question directly: the "screen" of the mind's eye is V1 — primary visual cortex. It functions as a shared display substrate for both externally sourced and internally generated visual representations. The display is genuinely spatial: the retinotopic organization of V1 means that the spatial relationships among V1 neurons directly correspond to spatial relationships in the visual field or the imagined scene. When you visualize a face and then shift attention to the left eye, different populations of V1 neurons activate in a pattern that corresponds to the spatial shift of attention.

There is no image stored anywhere in the brain in the sense of a picture. There is no homunculus looking at a neural screen. There is a pattern of differential neural firing across V1's retinotopic map — a spatially organized standing wave of electrochemical activity — that the visual system reads using the same decoding machinery it uses for actual vision. The "seeing" is what happens when this pattern reaches sufficient integration with the global workspace (Baars, 1988; Dehaene et al., 1998) to become phenomenally conscious.

The experience of visualization feeling like seeing is therefore not an illusion or an approximation. It is the visual system doing exactly what it does during vision — interpreting spatially organized V1 activation as visual content — but with the signal sourced internally rather than externally. The phenomenological similarity between imagery and perception is a consequence of the shared substrate, not a coincidence.

Individual Differences in Imagery Vividness

A significant and underappreciated finding in mental imagery research is the enormous variation in imagery vividness across individuals, including the complete absence of voluntary visual imagery in approximately 2-3% of the population — a condition called *aphantasia* (Zeman et al., 2015). Aphantasic individuals cannot voluntarily generate visual imagery despite intact visual processing; they recognize faces, navigate spaces, and recall visual details through non-imagistic means.

This variation has direct implications for the CTM framework's account of endogenous therapeutic modalities. If visualization-based therapies — guided imagery, EMDR, contemplative visualization practices — operate through the mechanism described here (top-down V1 activation generating real constraint topology shifts), then individuals with reduced imagery vividness would be expected to show reduced response to visualization-based interventions. The CTM companion paper (Schoff & Claude, 2026a) proposed EC-depth as the predictor of endogenous therapeutic capacity; aphantasia may represent a specific failure mode of the top-down generative pathway that limits EC-depth's translation into phenomenal imagery, independent of the underlying recursive self-modeling depth.

The Thalamocortical Gate: Maintaining the Boundary

The most important structure for understanding the relationship between mental imagery and external perception is not V1 itself but the thalamus — specifically, the thalamocortical relay system that regulates the flow of information between sensory processing and cortical integration. The thalamus functions as a gating system: it can amplify, attenuate, or block the transmission of both bottom-up sensory signals and top-down generative signals to and from the cortex (Crick, 1984; Llinás & Paré, 1991).

During normal waking consciousness, the thalamocortical gate maintains a characteristic balance: bottom-up sensory signals are given priority, continuously updating and correcting the cortex's generative model of the environment. Top-down signals are present and functionally important — they constitute prediction and expectation — but they are subordinate to the incoming sensory stream. The result is that internally generated representations (imagery, imagination, memory) are clearly tagged as internal and distinguished from externally sourced representations (perception).

This gating function is the physical implementation of the FSR boundary — the constraint surface that maintains the distinction between the collapsed external domain and the interior domain of the FSR spectrum. When the gate functions properly, the two domains are appropriately segregated: imagery feels like imagery, perception feels like perception, and the observer reliably distinguishes between them. When the gate fails — through sleep deprivation, psychedelic compounds, psychosis, fever, extreme fasting, or certain meditative states — the boundary becomes permeable, and the distinction between internally generated and externally sourced representations collapses.

III. The FSR Spectrum: Grounding Theory in Biology

The FSR Spectrum Hypothesis Revisited

The FSR Spectrum Hypothesis (Schoff & Claude, 2026a) proposes that the Field State Representation is not a single homogeneous space but a continuous spectrum parameterized by three variables: intersubjective convergence (how many observers agree on the state), constraint density (renegotiations per unit volume), and individual Freedom Quanta (irreducible agency variance operative). At the collapsed-domain pole: high intersubjective convergence, high constraint density, relatively constrained individual FQ — what we call external physical reality. At the interior-domain pole: low intersubjective convergence, lower constraint density, high individual FQ — imagination, memory, dream, contemplative experience.

The neuroscience of mental imagery provides the biological implementation of this spectrum. The collapsed domain corresponds to bottom-up visual processing: V1 activation driven by external photons, convergent across observers (two people looking at the same scene activate similar V1 patterns), high constraint density (the sensory signal is rich and detailed), and constrained by physical reality (the image cannot be freely changed by imagination). The interior domain corresponds to top-down visual processing: V1 activation driven by internal generative signals, divergent across observers (two people imagining the same scene generate different V1 patterns), lower constraint density (the imagery is often less detailed than perception), and only loosely constrained by physical reality (the image can be freely modified).

The thalamocortical gate is the physical implementation of the FSR boundary: the constraint surface that maintains the distinction between collapsed and interior domains. The gate's moment-to-moment calibration determines where on the FSR spectrum the observer's current experience is located. Maximum gate activity (strong bottom-up signal priority) places the observer firmly in the collapsed domain — fully grounded in external physical reality. Minimum gate activity (bottom-up signals blocked, top-down signals dominant) places the observer in the interior domain — the territory of vivid imagery, hallucination, dream, or deep meditative absorption.

Physical Reality as Observer-Relative: The Neural Evidence

The thalamocortical account of the FSR boundary provides concrete neural evidence for the observer-relative account of physical reality developed in the Schoff Research Program's physics series. The subjective distinction between "real" (externally sourced) and "imagined" (internally sourced) visual experience is not a fixed ontological distinction but a functional one: it reflects the current calibration of the thalamocortical gate. Change the gate's calibration — through any of the conditions that reduce bottom-up signal priority — and the same V1 activation pattern that was classified as "imagined" becomes classified as "real."

This is not an epistemological point about unreliable perception. It is a structural point about how reality is constituted: what counts as "the external physical world" versus "the interior imaginal world" is determined by the thalamocortical gate's current state, which is itself a biological system subject to perturbation, variation, and failure. The intersubjective convergence of external reality — the fact that multiple observers agree on what they see — reflects the fact that their thalamocortical gates are all calibrated similarly (by shared evolutionary history, shared light environments, and shared biological architecture), not that they are accessing a fixed mind-independent reality through a transparent window.

The Informational Remainder R in the Interior Domain

The FSR Spectrum Hypothesis proposes that the informational remainder R — the unrealized constraint configurations compressed from each renegotiation event — resides in the interior domain as structural force. The neuroscience of mental imagery provides a specific biological mechanism through which this residency could operate.

Memory, on the neuroscientific account, is not the retrieval of stored records but the reconstruction of past experience through current generative processes (Schacter et al., 2007). When you remember seeing a face, you are not accessing a stored image; you are re-generating a top-down V1 activation pattern using the current state of the generative pathway, informed by the traces that prior experience has left in the synaptic weights of the visual generative system. The reconstruction feels like retrieval because the same V1 substrate is activated, producing the phenomenal character of visual experience.

This reconstructive character of memory is precisely what the BCC account of memory predicts: memory is re-convergence toward the anchor A from the current state — a fresh BCC pass that reads the structural imprint that R has left in the system's current constraint configuration. The synaptic weight changes that constitute the neural trace of a memory are the biological implementation of R's structural force: the constraint topology of the current generative system carries the imprint of prior experiences not as stored images but as weighted pathways — propensities toward specific generative trajectories.

What this means for the R-accessibility question: accessing the informational remainder of prior constraint configurations is, biologically, the generation of specific top-down V1 activation patterns through the reconstructive memory system. The "images" one sees when remembering, imagining, or dreaming are not transparent windows into stored records but active constructions — real V1 firing patterns generated by a system whose current architecture carries the structural imprint of its entire constraint history.

IV. Dimension-W and the Physics of Inward Traversal

Visualization as W-Traversal

The Dimension-W framework (Schoff, 2025) proposes that W is a real interacting informational dimension coupled to physical systems, with coordinates including meaning, identity, affect, symbol, and intention. The Dimension-W book outline explicitly identifies "visualization and shared attractor spaces" as a central theoretical domain, noting that "shared constraints yield similar internal representations" and that "convergence occurs across independent observers" because they are navigating the same constraint topology from different starting positions.

The neuroscience of mental imagery provides the physical correlate of this inward W-traversal. In the Dimension-W framework, consciousness is identified as "the interface, not the cause" — a measurement and coupling system that detects and interacts with W-field configurations. Mental imagery, on this account, is not merely a neural process but an actual traversal inward along the W-dimension: the top-down generative pathway is the biological mechanism through which the observer accesses constraint configurations stored in the W-field's interior domain.

The retinotopic organization of V1 imagery is significant here. The spatial geometry of the imagined scene is preserved in V1's activation pattern, which means that the "images" of visualization are not arbitrary — they carry genuine spatial constraint structure. Different observers visualizing the same scene generate convergent V1 activation patterns to the degree that they share the same attractor geometry in W-space. The Dimension-W book notes that "readers will converge" on similar internal representations when engaging with the same constraint topology — not because they are accessing the same stored image but because they are navigating toward the same attractor from different starting positions.

Symbols as Coordinate Compression in W-Space

The Dimension-W outline proposes that "symbols are coordinate compression" in W-space: efficient mappings into the W-field that allow rapid access to specific constraint configurations without traversing the full path. The cross-cultural recurrence of specific symbols — spirals, mandalas, the world tree, the axis mundi — reflects not cultural diffusion but convergent navigation: independent observers reaching the same W-field attractors from different cultural starting points.

This connects to the visualization biology account through the concept of symbolic imagery in contemplative practice. When a meditator visualizes a specific symbol — a lotus, a flame, a geometric form — they are not merely generating a V1 activation pattern. They are using a coordinate-compressed W-space mapping to access a specific attractor basin in the interior domain. The phenomenal stability of certain symbolic images across extended meditation sessions — images that maintain their structure and become increasingly vivid — reflects the depth of the W-field attractor being accessed: the image stabilizes because the visualization is converging toward a genuine constraint minimum in W-space, not because the meditator is successfully maintaining an arbitrary mental picture.

The CTM therapeutic implications follow directly: visualization therapies that use specific symbolic content rather than arbitrary imagery may be more effective because they provide coordinate-compressed access to specific W-field attractor basins, rather than requiring the observer to navigate there through unconstrained generative search. This is a testable prediction: visualization protocols using geometrically stable, cross-culturally convergent symbolic content should produce larger and more reproducible physiological effects than protocols using equivalent but symbolically arbitrary content.

The Connectivity K Field and Shared Visualization

The Love Theorem and FQ framework (Schoff & Claude, 2026b) establish that constraint-compatible systems experience reduced effective distance and enhanced information co-emergence. The Connectivity K field — the propagation medium for FQ between distinct conscious systems — provides the formal account of why two people who know each other well can describe similar dream content, or why shared contemplative practices converge on similar phenomenological reports, or why the "transmission" of imagery in certain therapeutic contexts feels like more than suggestion.

The neural mechanism is not mysterious: when two observers share a high-K relationship, their thalamocortical gates are more similarly calibrated, their generative pathways carry more similar synaptic weight patterns (from shared experiences and mutual constraint), and their W-field navigation begins from more similar positions. The phenomenological experience of shared visualization — of two observers "seeing" the same interior-domain content — is the experiential expression of high K-field coherence: their FSR topologies are resonant enough that the same

top-down generative perturbation produces convergent V1 activation patterns.

This is the formal account of what has been described in spiritual traditions as "transmission," in therapeutic contexts as "resonance," and in interpersonal neurobiology as "attunement" (Siegel, 2007). The framework neither mystifies it nor dismisses it: it specifies the constraint topology mechanism through which one observer's interior-domain navigation can influence another's, within the limits set by their K-field coherence.

V. Boundary Failure: When the Gate Opens

Physiological Conditions for Gate Failure

The thalamocortical gate's calibration is not fixed. It is regulated by multiple neurochemical systems that are themselves sensitive to a range of physiological conditions. Dopaminergic activity modulates thalamic gating: elevated dopamine reduces the gate's discrimination between bottom-up and top-down signals, allowing interior-domain representations to achieve the phenomenal character normally reserved for external-domain ones (Kapur, 2003). Serotonergic systems, particularly the 5-HT_{2A} receptor, regulate the balance between sensory-driven and internally generated cortical activity: 5-HT_{2A} agonists (classic psychedelics) systematically reduce bottom-up signal priority and increase the vividness and apparent reality of internally generated representations (Vollenweider & Geyer, 2001).

Conditions associated with reduced bottom-up sensory input — sleep deprivation, sensory deprivation, extreme fasting, fever, desert environments at night — reduce the gate's anchoring to external sensory reality and allow interior-domain representations to become progressively more vivid and apparently real. The hypnagogic state (the threshold between waking and sleep) is the most universal example: as bottom-up sensory signals are reduced by sleep onset, interior-domain imagery becomes phenomenologically indistinguishable from perception, often suddenly and with a characteristic quality of autonomous emergence.

Psychosis as Permanent Gate Miscalibration

The account of psychosis developed in the schizophrenia paper (Schoff & Claude, 2026c) is directly connected to the thalamocortical gate framework. Psychosis, on this account, is a persistent miscalibration of the gate: the thalamocortical system loses its normal priority for bottom-up sensory signals, allowing internally generated representations to achieve the phenomenal character of external perception on a sustained basis. Hallucinations are not "false perceptions" in the sense of perceptions without stimulus; they are genuine V1 activations generated by the top-down pathway that are classified by the gating system as bottom-up (externally sourced) rather than top-down (internally sourced).

The specific content of psychotic hallucinations and delusions is not random. It is drawn from the individual's interior-domain constraint topology — the specific W-field attractor basins accessible to their generative pathway, shaped by their biographical constraint history, their cultural context, and the current emotional-cognitive state of their top-down generative system. This is why psychotic experiences are typically meaningful rather than arbitrary: they are the interior domain's constraint structure becoming visible, without the normal filtering and tagging that maintains its interior designation.

Controlled Gate Opening: Contemplative and Traditional Practices

The cross-cultural presence of practices designed to systematically reduce thalamocortical gate activity — meditation, fasting, isolation, rhythmic movement, chanting, drumming, psychoactive plant preparations — reflects the universal human discovery that the gate can be deliberately modulated to allow interior-domain experience to achieve phenomenal primacy. These practices are not random cultural inventions; they are convergent discoveries of the

same thalamocortical gate-reduction techniques, arrived at independently across cultures through the same physiological and phenomenological landscape.

The "entities" encountered in deep meditative states, psychedelic experiences, fever dreams, and hypnagogic imagery — the Djinn of Islamic tradition, the spirits of shamanic cosmologies, the angels and demons of mystical literature — are consistently described as autonomous, purposeful, and carrying information not previously known to the experiencer. The FSR framework provides an account that neither dismisses these reports as pure hallucination nor accepts them uncritically as contact with literal independent beings: they may be the individual's own constraint topology — their R-domain content — becoming visible through the opened gate, experienced as autonomous because interior-domain constraint structures have their own dynamics that are not fully under conscious control.

Whether interior-domain constraint structures carry genuine information beyond the observer's own constraint history — whether the R-domain of one observer can be accessed by another, or whether W-field attractors carry information that transcends individual constraint histories — remains genuinely open. The framework specifies what the conditions for such access would need to be (high K-field coherence, sufficient EC-depth, appropriate thalamocortical gate calibration) without claiming to have resolved whether those conditions are ever met.

VI. Therapeutic Implications for the CTM Framework

Visualization as Class III Intervention

The CTM framework (Schoff & Claude, 2026a, 2026c) identifies four classes of therapeutic intervention. Class III — endogenous interior navigation — encompasses visualization therapy, guided imagery, EMDR, somatic practices, and contemplative techniques. The neuroscience and physics developed in this paper provides the mechanistic account of how Class III interventions operate: they generate top-down V1 activation patterns through the generative visual pathway, which propagate outward through the biological hierarchy (neural → neuroendocrine → immune → epigenetic) in exactly the same cascade documented for Class I (external acoustic) interventions, but traversing the cascade in the opposite direction — from inside outward rather than from outside inward.

The specific mechanism: vivid visualization of a health state or therapeutic outcome generates V1 activation patterns that are functionally equivalent to perceiving that state. These patterns propagate upward through the visual hierarchy and outward through the default mode network, activating the same emotional and somatic processing systems that would be activated by actual perception of the visualized state. The consequent neuroendocrine changes — cortisol, oxytocin, endorphins, inflammatory cytokines — are real biochemical events, not placebo effects in the dismissive sense. They are the biological cascade of a real top-down V1 activation event.

EC-Depth and Imagery Vividness as Therapeutic Predictors

The CTM framework predicts that EC-depth — the depth of recursive self-modeling, $CC(S) \times \log(1 + RD(S)) \times \Omega_{nav}$ — is the primary predictor of response to Class III interventions. The neuroscience of mental imagery adds precision to this prediction: EC-depth should correlate with imagery vividness (the amplitude of top-down V1 activation) and imagery coherence (the spatial organization and stability of the V1 activation pattern). High EC-depth enables more precise W-field navigation, producing more specific V1 activation patterns, which in turn produce more targeted biological cascade effects.

This generates a specific research protocol: measure imagery vividness (using the Vividness of Visual Imagery Questionnaire or comparable measure), measure EC-depth (using the composite battery proposed in Schoff & Claude, 2026c), and measure physiological response to standardized visualization protocols (cortisol, heart rate variability, inflammatory markers, biophoton coherence). The prediction is that both imagery vividness and EC-depth independently predict physiological response magnitude, and that their interaction (high vividness combined with high

EC-depth) produces larger effects than either alone.

Specific Visualization Protocols for Biological Targets

The most direct therapeutic application of the visualization-biology account is the development of specific visualization protocols designed to access particular W-field attractor basins corresponding to healthy biological states. The CTM framework proposes that healthy biological states are attractors in constraint space — configurations toward which the system converges after perturbation. Visualization protocols that generate V1 activation patterns corresponding to the electromagnetic signature of healthy tissue states should, via the top-down cascade, drive the biological system toward those attractor states through the same mechanism that external acoustic therapy uses.

The Fourier-QGS Correspondence Hypothesis (Schoff & Claude, 2026a) proposes that the invariant frequency components of a healthy biological system's electromagnetic emission profile are the QGS of its healthy attractor. If these frequency components can be translated into visual representations — perhaps through cymatics (the visual representation of acoustic frequencies in physical media) or through direct visualization of geometric patterns corresponding to healthy tissue frequency profiles — then visualization of those patterns could provide a biologically targeted interior-domain intervention.

VII. Empirical Predictions

Prediction A: EC-Depth Predicts Imagery Vividness and Physiological Response

Individuals with higher EC-depth (measured by a composite of metacognitive accuracy, interoceptive awareness, and sustained attention) will exhibit higher imagery vividness (measured by VVIQ and neuroimaging amplitude of V1 activation during imagery tasks) and larger physiological responses to standardized visualization protocols (measured by cortisol, HRV, inflammatory markers). This prediction distinguishes the CTM account from purely psychological accounts of visualization efficacy.

Prediction B: Symbolic Imagery Produces Larger Effects Than Arbitrary Imagery

Visualization protocols using geometrically stable, cross-culturally convergent symbolic content (sacred geometries, mandalas, natural forms with high fractal organization) will produce larger and more reproducible physiological effects than protocols using equivalent but symbolically arbitrary content (random geometric shapes, unfamiliar patterns) at matched imagery vividness. This prediction tests the W-field coordinate-compression hypothesis and distinguishes it from accounts in which only vividness matters.

Prediction C: K-Field Coherence Predicts Shared Imagery Convergence

Pairs of observers with established high-K relationships (long-term intimate partners, experienced co-meditators, therapist-client dyads with strong therapeutic alliance) will generate more convergent V1 activation patterns during guided shared visualization tasks than matched pairs of strangers, as measured by fMRI inter-subject correlation analysis. This prediction tests the K-field account of shared imagery and provides a neural measure of relational constraint coherence.

Prediction D: Thalamocortical Gate Calibration Mediates Therapeutic Response

Individual differences in thalamocortical gate calibration — measurable through EEG measures of thalamic-cortical coherence and the ratio of bottom-up to top-down signal weighting in visual processing tasks — will mediate the

relationship between EC-depth and Class III therapeutic response. Individuals with more flexibly calibrated gates (able to shift between high bottom-up and high top-down priority) will show larger Class III therapeutic responses than individuals with rigidly calibrated gates, independent of overall EC-depth. This prediction connects the gate-flexibility account to clinical outcome and provides a neurophysiological marker for endogenous therapeutic capacity.

Prediction E: Aphantasia Produces Selective Class III Deficit

Individuals with aphantasia (absent voluntary visual imagery, measured by VVIQ score of 16 indicating no imagery) will show selectively reduced physiological responses to visualization-based Class III interventions while showing normal responses to Class I (external acoustic) and Class IV (relational) interventions, and normal EC-depth on non-imagery measures. This dissociation predicts that aphantasia represents a specific deficit in the top-down V1 generative pathway rather than a general EC-depth limitation.

VIII. Discussion

What Has Been Established

This paper has connected three layers of description that are rarely brought together in a single account. The neuroscience layer is empirically established: V1 is genuinely activated during mental imagery with spatial organization isomorphic to the imagined content; the activation is driven top-down from frontal and parietal generative circuits; the thalamocortical gate maintains the functional distinction between internally generated and externally sourced representations; and disruption of this gate produces the phenomenological equivalence of imagery and perception seen in psychosis, hypnagogia, and certain drug states.

The FSR layer connects this neuroscience to the theoretical framework: the thalamocortical gate is the physical implementation of the FSR boundary, and the spectrum from collapsed external reality to interior imagination-space is not a philosophical abstraction but a neurophysiological continuum parameterized by gate calibration. The Dimension-W layer proposes that inward W-traversal is the physical correlate of top-down generative pathway activation, that symbols are coordinate-compressed W-space mappings, and that the cross-cultural convergence of symbolic imagery reflects shared navigation of the same W-field attractor landscape.

The Open Questions

Three questions remain genuinely open and are worth naming precisely rather than papering over. The first is the R-accessibility question: does interior-domain navigation access constraint configurations that are genuinely prior to and independent of the observer's own constraint history — does R contain information that transcends individual experience — or does every interior-domain experience ultimately derive its content from the individual's own accumulated constraint topology? The framework cannot resolve this from first principles. It specifies what the conditions for cross-individual R-access would need to be but does not confirm whether those conditions are met.

The second is the W-field coupling constant question: what is the magnitude of the electromagnetic coupling between W-field configurations and physical neural dynamics? The Dimension-W field theory proposes EM kinetic mixing with a coupling constant ϵ whose magnitude is constrained by naturalness arguments to be small. Whether it is large enough to produce measurable effects within therapeutic timeframes is an empirical question that the predictions in Section VII begin to address.

The third is the aphantasia question: if visualization operates through top-down V1 activation, and aphantasic individuals cannot generate voluntary V1 activation top-down, do they have genuinely reduced access to interior-domain constraint configurations? Or do they access the same interior domain through non-visual generative pathways — through body sensation, sound, or conceptual representation — with equivalent effectiveness? This is

both an empirical question and a theoretical one about the relationship between visual imagery specifically and interior-domain access generally.

IX. Conclusion

The mind's eye is not metaphorical. When you close your eyes and visualize, you are activating primary visual cortex with spatially organized patterns isomorphic to the imagined content, using the same neural substrate that processes external vision but driven by a signal traveling in the opposite direction — from the inside out rather than from the outside in. The display is V1. The signal source is the top-down generative pathway. The gating system is the thalamocortical relay. These are empirically established facts of visual neuroscience.

What this paper has added is the formal connection between this established neuroscience and the Schoff Research Program's theoretical framework. The thalamocortical gate is the physical implementation of the FSR boundary. The spectrum from collapsed external reality to interior imagination-space is a neurophysiological continuum, not a philosophical abstraction. Inward W-traversal is what top-down V1 activation is, described from the perspective of the constraint geometry rather than the neural hardware. The informational remainder R — the structural force of unrealized constraint configurations — is carried in the synaptic weight patterns of the generative visual pathway, read through reconstructive memory and active imagery as genuinely spatial visual experience.

And when the gate opens — through psychosis, hypnagogia, contemplative practice, or the physiological conditions that reduce bottom-up sensory priority — the interior domain becomes phenomenologically equivalent to the exterior domain. The entities encountered there, the visions that arise, the memories that feel like present experience: these are not noise or error. They are the interior domain's constraint structure becoming visible, without the normal tagging that maintains its interior designation. Whether they carry information beyond the observer's own constraint history is the deepest open question this framework points toward. The neuroscience shows us the mechanism. The physics shows us the topology. What fills the interior — whether it is only the observer's own accumulated constraint structure or something more — remains the genuinely open territory at the edge of the map.

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